

Name: _____
Class: _____
Date: _____

Mr. Rawson • GCSE Physics

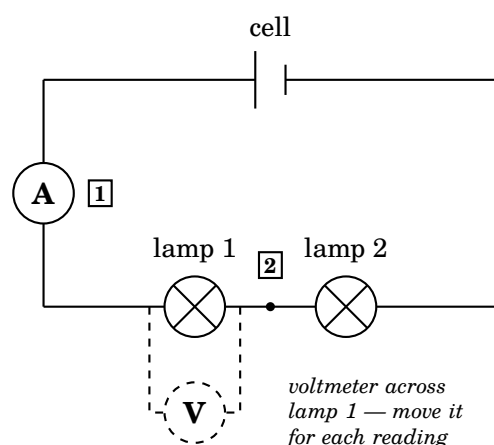
Building Series and Parallel Circuits

Key concept

There are only two ways of joining components: **series** — one single loop, so the charge has no choice of route — and **parallel** — the circuit branches at a junction, so the charge has more than one route. Whatever you measure today, $V = IR$ still applies to every single component.

Meters: the **ammeter** goes **in the loop** (in series); the **voltmeter** goes **across a component** (in parallel with it). Get these the wrong way round and the circuit will not work.

Circuit 1 — series



Method

1. Build the circuit shown. Ask before you connect the supply.
2. Record the current at position 1.
3. Move the ammeter to position 2 (between the lamps) and record again.
4. Connect the voltmeter *across* lamp 1, then lamp 2, then the cell, recording each potential difference.
5. Unscrew one lamp. Watch the other.

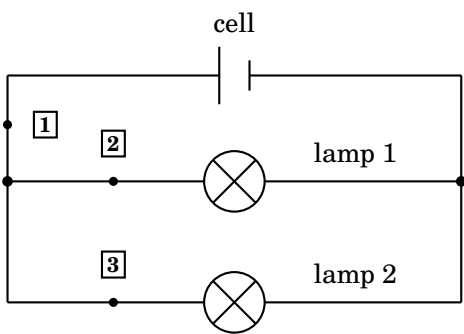
Position	Current / A
1	
2	

Voltmeter across	pd / V
lamp 1	
lamp 2	
the cell	

1) **State** what you notice about the current at different points in a series circuit. [1]

2) **Compare** the pd across the two lamps with the pd across the cell. [2]

Circuit 2 — parallel



Method

- 1. Rebuild to this diagram: the two lamps are now on separate *branches*.
- 2. Put the ammeter in at position **1** (before the junction) and record. Repeat at positions **2** and **3**, one at a time.
- 3. Connect the voltmeter across lamp 1, then lamp 2, then the cell.
- 4. Unscrew one lamp. Watch the other.

Position	Current / A	Voltmeter across	pd / V
1		lamp 1	
2		lamp 2	
3		the cell	

3) **State** how the currents in the two branches (positions 2 and 3) relate to the current from the cell (position 1). [1]

4) **State** what you notice about the pd across each lamp in the parallel circuit. [1]

5) **Explain** what happened to the other lamp when you unscrewed one — in the series circuit *and* in the parallel circuit. [2]

The rules

	Series	Parallel
Current		
Potential difference		
Adding another resistor		

6) **Calculate** the resistance of lamp 1 in your *series* circuit, using your own measured current and potential difference. [3]

Careful: V must be the pd across *that lamp alone* — not the pd across the cell.